Smart AutoRoof Weather Monitoring System

Using ESP8266

An Embedded System Project.

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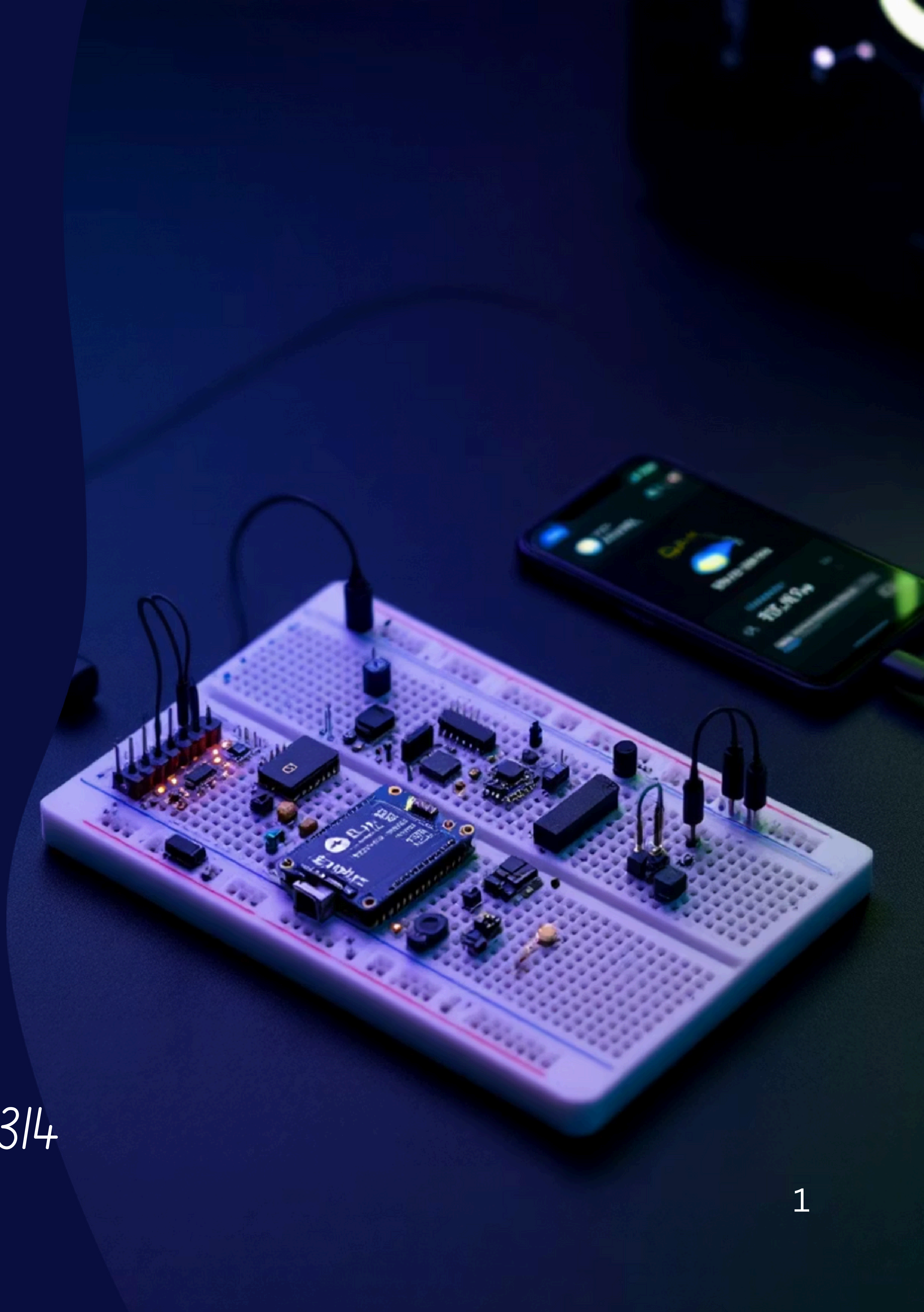

Presentation Outline

01

Introduction & Objectives

Understanding the project's scope and goals.

02

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Exploring existing IoT weather solutions.

03

Hardware Impllimentation

Circuit Diagram,Circuit connections.

Detailed look at hardware used.

04

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Steps taken during implementation.

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Results & Discussion

Analysis of system performance.

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Conclusion & Future Work

Summary and potential enhancements.

This presentation will guide you through the development of a smart weather forecasting and automation system, from conception to practical application.

Introduction: Real-time Weather Monitoring and Automation

This IoT-based project is designed to monitor multiple crucial weather parameters in real-time. By leveraging a network of sensors, it continuously collects data on environmental conditions, offering a comprehensive overview of the local weather.

- Temperature
- Humidity
- Air Pressure
- Rainfall
- Light Intensity

All collected data is seamlessly transmitted to the Blynk IoT platform, providing users with instant access to live weather updates via their smartphones. Beyond monitoring, the system also incorporates an automation feature, controlling a servo motor based on specific weather conditions.

Literature Review: Advancements in IoT Weather Solutions

IoT-based Weather Stations

Widely adopted in smart farming for optimized irrigation and crop management, and in various automation applications for proactive environmental control.

Blynk Platform Integration

A powerful IoT platform enabling easy cloud monitoring and control via smartphones, facilitating intuitive user interaction with sensor data.

Sensor-based Automation

Enhances safety and operational efficiency by allowing systems to respond autonomously to environmental changes, reducing human intervention.

Servo Motor Applications

Crucial for automatic mechanical movements, commonly used in smart gates, retractable roofs, and automated shutters, providing precise positioning and control.

Project Objectives: Building an Intelligent Weather System

1

Precise Parameter Measurement

Utilize DHT11 for temperature and humidity, BMP180 for atmospheric pressure, a rain sensor for precipitation, and an LDR for light intensity.

2

Real-time Data Transmission to Blynk

Ensure seamless and continuous transfer of sensor data to the Blynk IoT platform for remote monitoring and visualization.

3

Rainfall-Triggered Automation

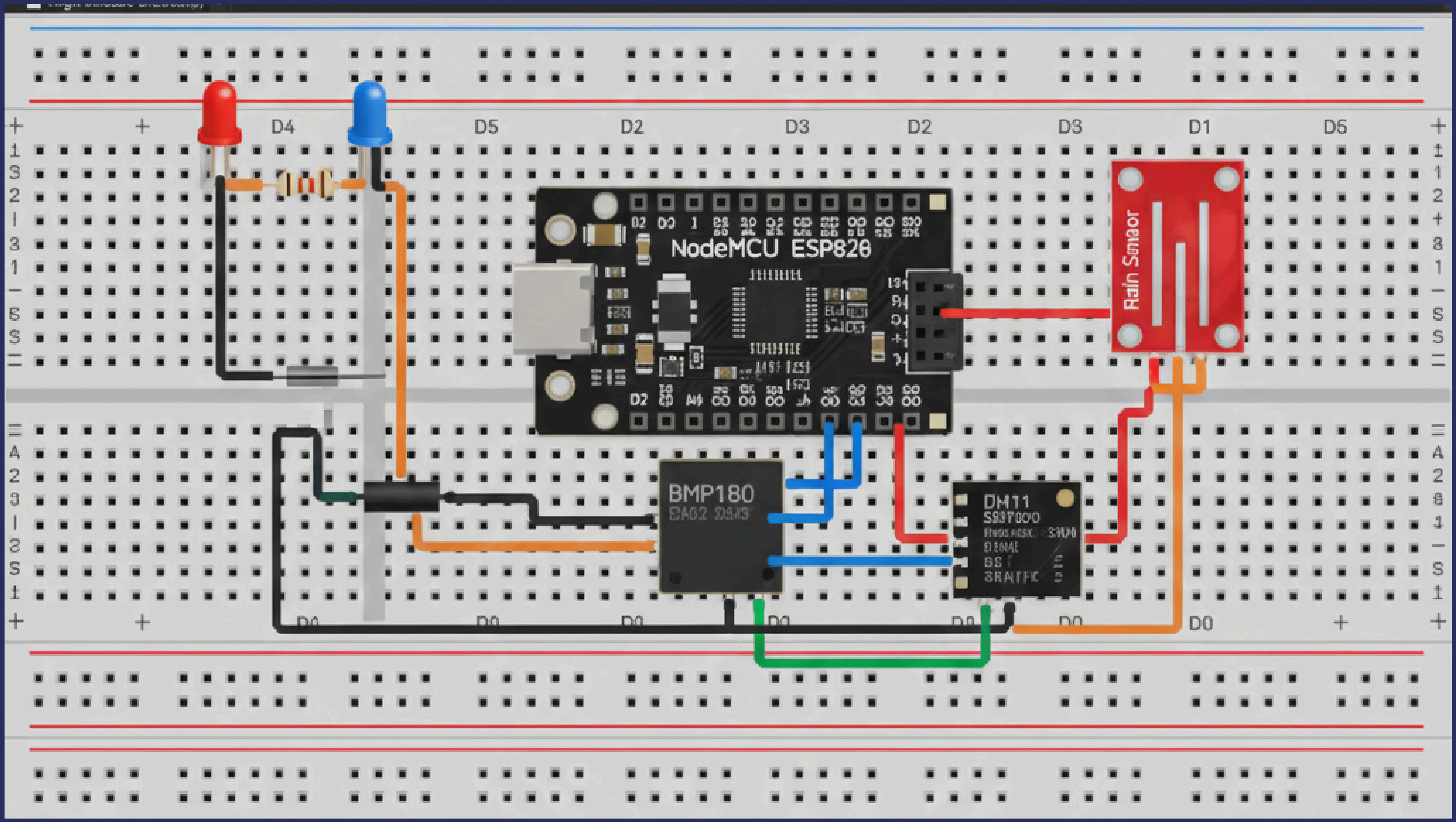
Implement an automatic mechanism using a servo motor to close a gate or roof upon detection of rainfall, enhancing protection and convenience.

4

Develop a Cost-Effective and Scalable Station

Design a weather station that is affordable to build and easily expandable for future functionalities and sensor integrations.

Weather monitoring System Circuit :



Weather monitoring System Circuit Connections:

Component	Pin on Component	Connect to NodeMCU
DHT11 Sensor	Data	D5
	VCC	3.3V
	GND	GND
Rain Sensor	A0	A0
	VCC	3.3V
	GND	GND

Component	Pin on Component	Connect to NodeMCU
BMP180	SDA	D2
	SCL	D1
	VCC	3.3V
	GND	GND
LDR Module	VCC	3.3V
	D0	D0
	GND	GND

Weather monitoring System Circuit Connections:

Component	Pin on Component	**Connect to NodeMCU**
Servo Motor	Signal	D6
	VCC	3.3V
	GND	GND
LED 1	Anode	D4
LED2	Anode	D7
Both LEDs	Cathode	GND

Core Components of Weather Monitoring System: Hardware Overview

ESP8266 NodeMCU

The brain of our IoT system, enabling Wi-Fi connectivity.

DHT11 Sensor

Measures ambient temperature and humidity levels.

BMP180 Sensor

Provides accurate atmospheric pressure and temperature data.

LDR

Measures ambient light intensity for day/night detection.

Servo Motor

Automates physical movements like closing a gate.

Rain Sensor

Detects precipitation and its intensity.

Component Spotlight: Detailed Specifications

ESP8266 NodeMCU

- Operating Voltage: 3.3V
- WiFi: 802.11 b/g/n
- Flash Memory: 4MB
- CPU: 80/160 MHz
- GPIO: 11 pins
- Analog Input: 1 (0–1V)
- Interfaces: UART, I2C, SPI

LDR Module:

- LDR: Detects ambient light, Fast response

DHT11 Sensor

- Temperature Range: 0–50°C
- Humidity Range: 20–90% RH
- Supply Voltage: 3.5–5.5V
- Output: Digital

BMP180 / BMP085 Sensor

- Pressure Range: 300–1100 hPa
- Temperature Accuracy:±1.0°C
- Voltage: 1.8–3.6V
- Interface: I2C

Rain Sensor

- Voltage Supply: 3.3–5V
- Outputs: Analog + Digital
- Adjustable Sensitivity
- Detects water drops/rainfall intensity

Servo Motor

- Servo Voltage: 4.8–6V
- Servo Angle: 0°–180°
- Servo Torque: 1.8 kg·cm
- Servo Control: PWM

Work Done: Implementation and Integration

Sensor Interfacing with NodeMCU:

Connected all sensors to the ESP8266 NodeMCU utilizing appropriate GPIO pins and communication protocols. Ensured secure physical wiring and verified initial sensor readings for data validity in the Arduino IDE environment.

Firmware Development:

Developed Arduino IDE code to read sensor data, process environmental parameters, and implement automatic logic for servo motor control based on rain sensor input. Incorporated debounce and filtering mechanisms to reduce noise and false triggers.

Cloud Integration with Blynk:

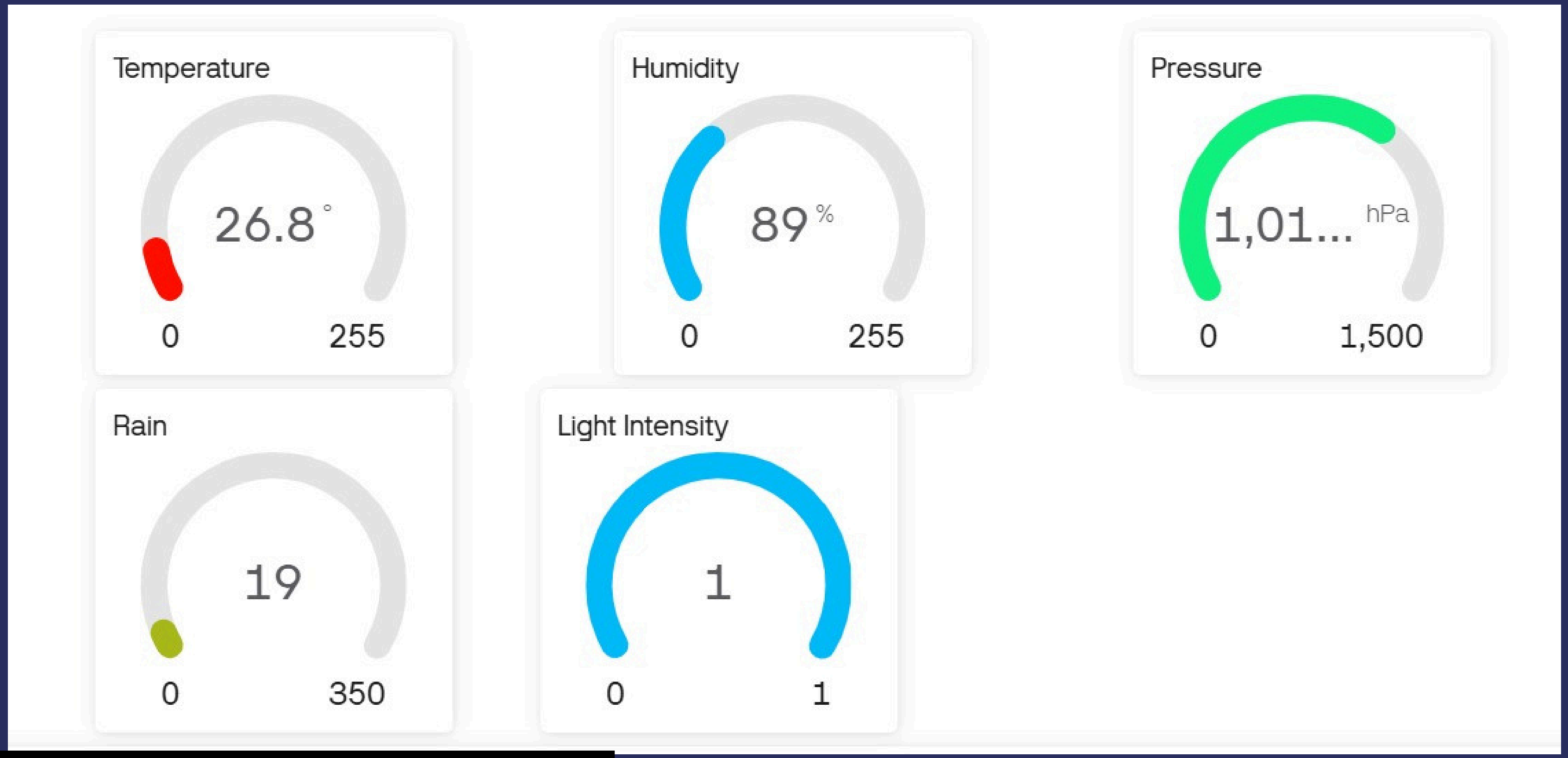
Configured Blynk widgets for displaying temperature, humidity, pressure, rainfall, and light intensity values in real time. Established communication via ESP8266's Wi-Fi module to transmit sensor data continuously to the cloud server.

Automation and Indicators:

Programmed the servo motor to close or open gate/roof automatically upon detecting rainfall intensity surpassing the threshold. Enabled LEDs to display system status, sensor activity, and automation state for instant visual feedback.

Results of Weather Monitoring System:

- *The system has successfully measured all the physical quantities in real time.*
- *All the sensor readings were accurately displayed on the Blynk IoT app without delay.*
- *The servo motor automatically rotates when rainfall hits the threshold 100 and roof will automatically covers the area .*
- *The system operated smoothly, showed stable readings, and met all the project objectives.*



Conclusion

Development of a Smart IoT Weather Station:

Successfully implemented a low-cost, scalable IoT-based weather forecast and automation system using ESP8266, multiple sensors, and Blynk cloud platform. Achieved real-time environmental data acquisition coupled with rain-triggered actuator control.

Enhancements in Safety and Monitoring:

Real-time monitoring and automation improve safety by protecting assets through automatic closure of roofs or gates during rainfall. Remote access via Blynk enhances user convenience and operational visibility.

Future Directions:

Potential upgrades include integrating AI-based weather predictions using machine learning models, adding solar power for energy independence, extending cloud analytics for long-term weather trends, and incorporating additional sensors such as soil moisture for agriculture-specific applications.

FUTURE ENHANCEMENTS

- **AI-based Weather Prediction:** Implement Machine Learning models for more accurate rainfall and temperature forecasting.
- **Automatic Irrigation:** Integrate soil moisture sensors to enable smart irrigation based on both soil conditions and rainfall data.
- **Solar-Powered Station:** Develop a self-sustaining system using solar panels for power in remote locations.
- **Smart Home Integration:** Expand automation capabilities to control other smart home devices based on weather parameters.
- **Cloud Analytics:** Utilize cloud storage and analytics for long-term trend analysis and predictive insights.

THANK YOU

Any Questions?